

Smart patch monitors diabetic wounds to prevent amputation risk

Researchers at KAIST in collaboration with the Korea Institute of Ceramic Engineering and Technology and Chungbuk National University have developed a self-regulating wearable OLED patch that combines light therapy and controlled



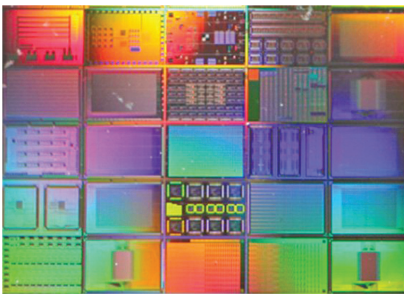
(From Left) Prof. Inkyu Park, Dr Seokjoo Cho, (Upper Pics, From Left) Prof. Ji-Hwan Ha, Researcher Jun-Ho Jeong, Prof. Wei Gao (Credit: www.kaist.ac.kr)

drug delivery, nearly doubling wound-healing speed in preclinical tests. The patch automatically adjusts antioxidant drug release in response to light intensity, helping maintain optimal treatment levels. Engineered by a team in Daejeon, South Korea, the device uses a skin-conform-

ing 630-nanometre OLED that delivers uniform illumination while remaining cool and stable during extended use. Mouse studies showed a 67% wound recovery rate after 14 days, compared with 35% in controls. The research was published in *Materials Horizons*.

Spintronic silicon chip advances AI

Engineered by a Japan-US research team, the chip was developed at Tohoku University in collaboration with the National Institute of Standards and Technology (NIST). Researchers have developed the world's first spintronic probabilistic bit



Photograph of test chips fabricated on a silicon substrate using semiconductor integrated circuit manufacturing processes (Credit: Shunsuke Fukami, William A. Borders et al)

(p-bit) integrated directly onto a silicon chip, a breakthrough that could enable more energy-efficient AI hardware and scalable probabilistic computing systems. The technology combines conventional silicon CMOS manufacturing with spintronic devices

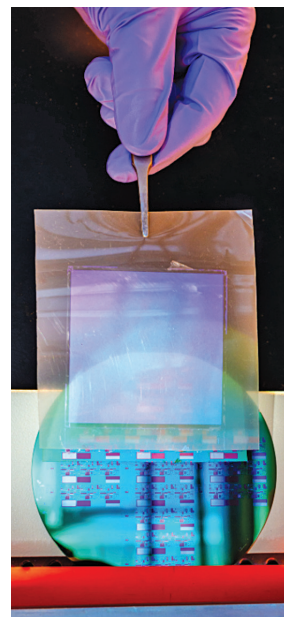
that generate intrinsic randomness, a key requirement for probabilistic computing. Fabricated using a 130-nanometre CMOS process, the device demonstrated both stochastic output fluctuations and controllable average output states. The advance brings spintronic p-computers closer to practical deployment for optimisation, inference, and machine-learning applications, offering a potential alternative to conventional deterministic processors as AI workloads continue to grow.

Crystal stress method reshapes nanochip manufacturing techniques

Research conducted at Rice University in Houston, Texas, demonstrated that alpha-molybdenum trioxide crystals can deform under an electron beam, generating highly ordered nanoscale wrinkles that function as optical gratings for guiding and manipulating light on chips. A new room-temperature chip fabrication technique uses stressed crystals to create nanoscale patterns on semiconductor-compatible materials, offering a simpler and potentially lower-cost approach to manufacturing photonic and optoelectronic devices. The method produces ultra-fine ripple structures on materials including silica, aluminium oxide, and silicon nitride without the high temperatures or complex processing required by conventional techniques. The team showed that pattern dimensions can be precisely controlled by varying crystal thickness and electron-beam intensity. By enabling single-step nanoscale patterning with minimal residue, the technology could support more efficient integration of optical functions into next-generation semiconductor and photonic systems.

Latest 3D chip stacking method could extend Moore's Law

Engineered by a team at the University of Illinois Grainger College of Engineering in Urbana-Champaign, the process uses ultrathin single-crystalline silicon nanomembranes transferred at temperatures below 200°C, avoiding heat damage to underlying circuits. Researchers have developed



A freestanding sheet of single-crystalline silicon nanomembrane is held above a silicon wafer patterned with its first layer of electronic circuits (Credit: <https://matse.illinois.edu>)

a low-temperature 3D chip manufacturing method that could extend Moore's Law by enabling silicon circuits to be stacked vertically without sacrificing performance. The technique allows multiple layers of high-performance silicon transistors to be integrated directly atop one another, increasing computing density while reducing power consumption and signal delays. The researchers successfully built three stacked silicon layers with up to 100% device yield and demonstrated functional logic and memory circuits. The scalable approach could support future processors, AI accelerators, and memory chips with significantly higher performance as conventional transistor miniaturisation approaches its physical limits.